

Category Intelligence

Retail product performance EDA — what the Category Manager should take into next week's supplier review

₹45.98 Cr

revenue, Jan–Jun 2026

107,836

transactions

12 stores

6 cities in Andhra Pradesh

25 products

in 5 categories

GROUP 10

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The problem we were asked to solve

CBIM Problem Canvas — completed before the analysis



Situation

A 12-store chain in Andhra Pradesh sells Electronics, Apparel, Grocery, Home & Kitchen and Personal Care. The quarterly supplier review is next week.



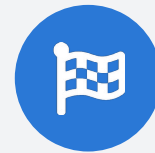
Complication

Six months of raw transactions, but no clear view of which categories and products have earned their shelf space — so suppliers would set the agenda.



Question

Which category do we back, which supplier do we challenge, and are discounts earning revenue or just giving away margin?



Success

The Category Manager walks in with one category to back, one supplier to challenge, a discount position and promotion timing — all with figures.

Stakeholder: the Category Manager **Single source of truth:** MUC01_Retail_Sales_Dataset.csv (1 Jan – 30 Jun 2026)

How we worked: six steps from raw data to a decision

Every step has code, a chart, and a plain-English business meaning

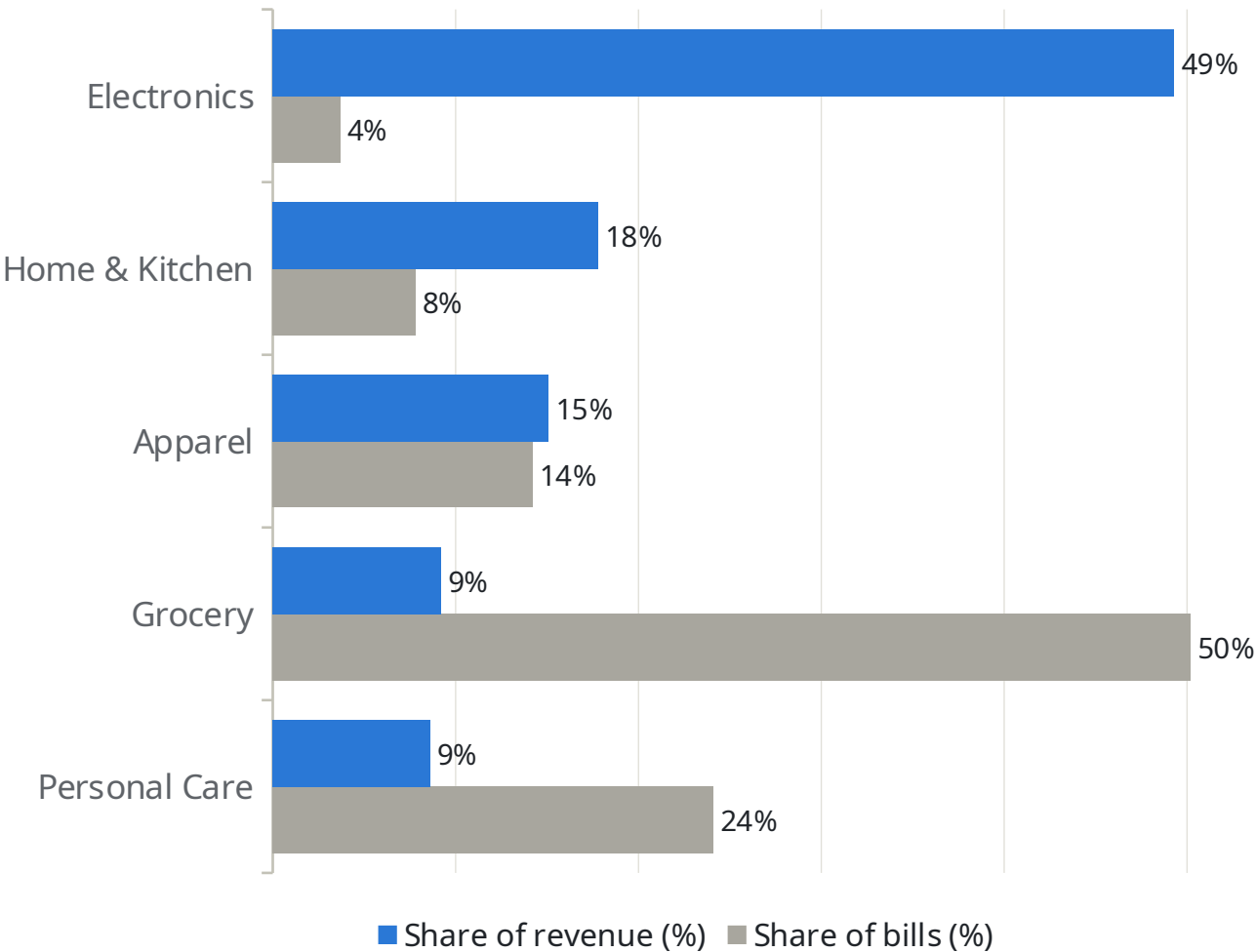


Design rules applied to every chart

- The title states the finding
- Blue = growing, red = declining, grey = stable — everywhere
- Money in ₹ Lakh / ₹ Crore; business labels, not column names
- Python, pandas, NumPy, Matplotlib, Seaborn
- Notebook re-run end-to-end from a fresh kernel — no errors
- Every figure in the text checked against the output

Electronics is 4% of bills but 49% of revenue

Grocery is the opposite — half of all bills, 9% of the money



₹56,420

average Electronics bill — one lost Electronics sale equals ~72 Grocery bills (₹781 each)

50%

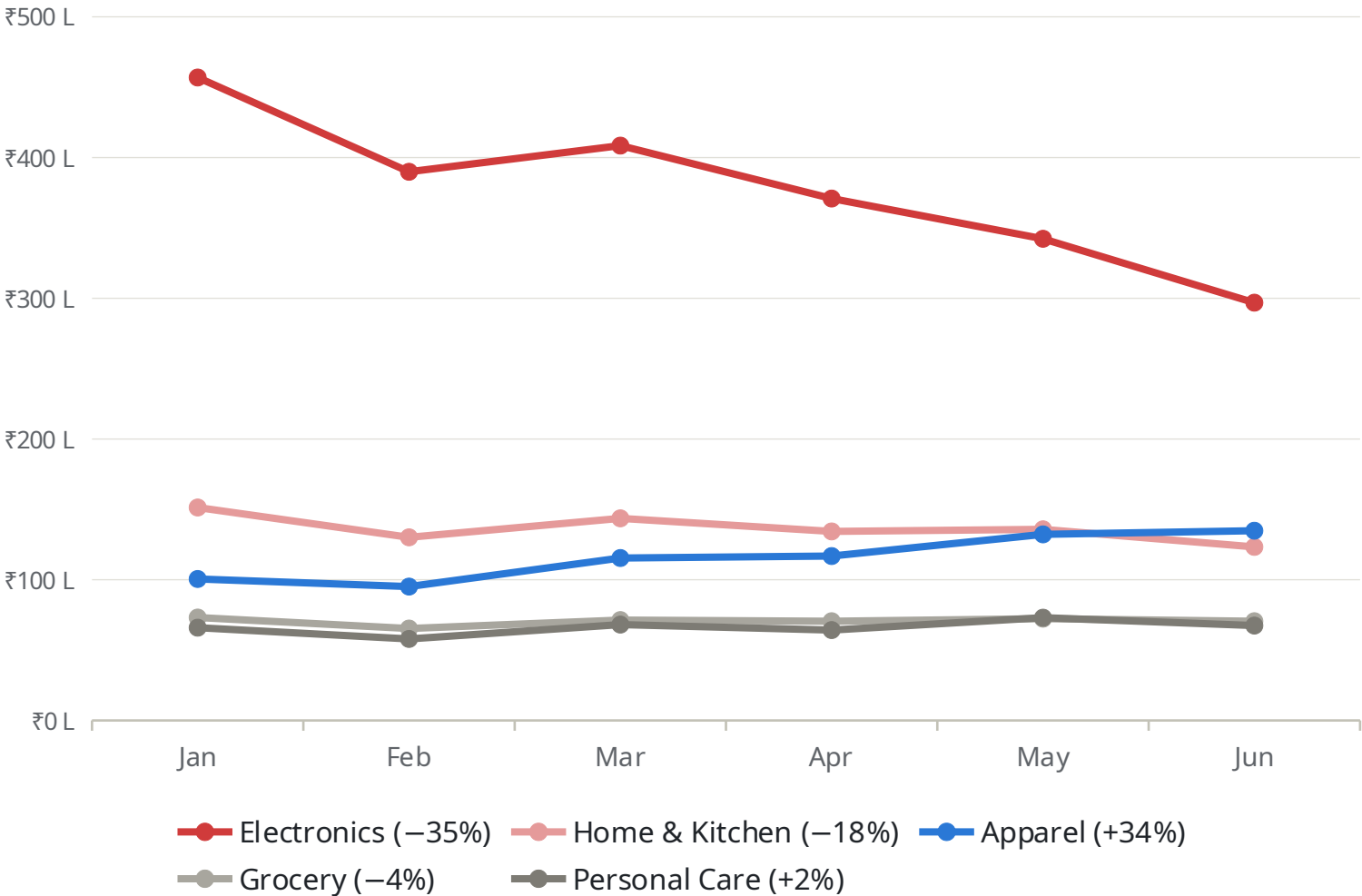
of all transactions are Grocery — its effect on customer visits is not measured here

₹4,264 vs ₹1,035

average vs median bill — a few big-ticket sales carry the chain

Electronics fell 35% while Apparel grew 34%

Monthly revenue by category (₹ Lakh), January–June 2026



–₹1.60 Cr

June Electronics revenue vs January (₹4.57 Cr → ₹2.97 Cr). Transactions –33%, prices –2%; the cause is not established by the data.

+34%

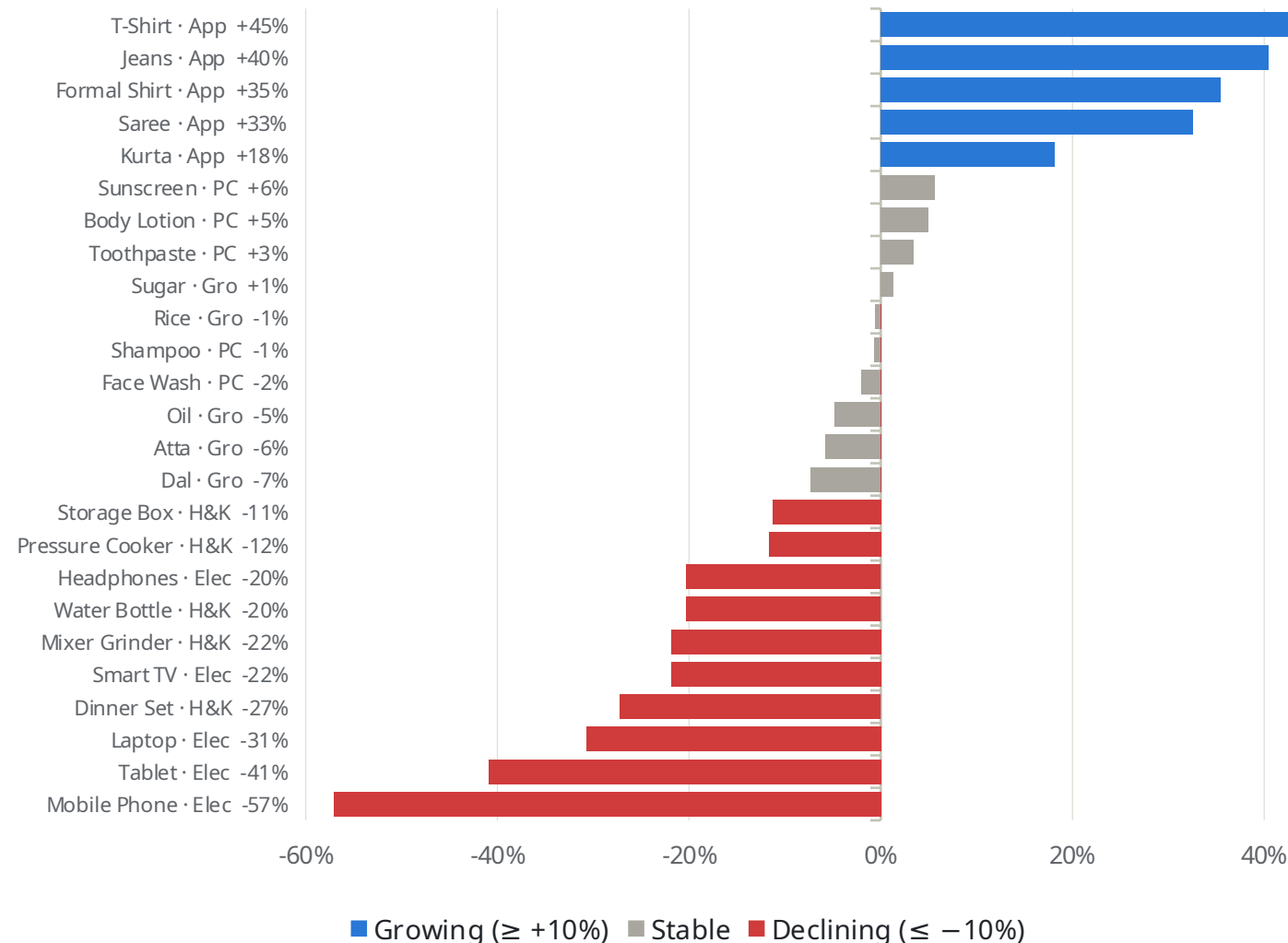
Apparel: ₹1.01 Cr → ₹1.35 Cr; recorded transactions +31% (customer and seasonal data needed to explain why).

+1.3%

The other four categories combined, excluding Electronics — Electronics accounts for the chain's –18% slide.

Mobile Phone fell 57% — the worst of 25 products

Revenue change January → June by product. Every Electronics product fell; every Apparel product grew.



The hidden problem

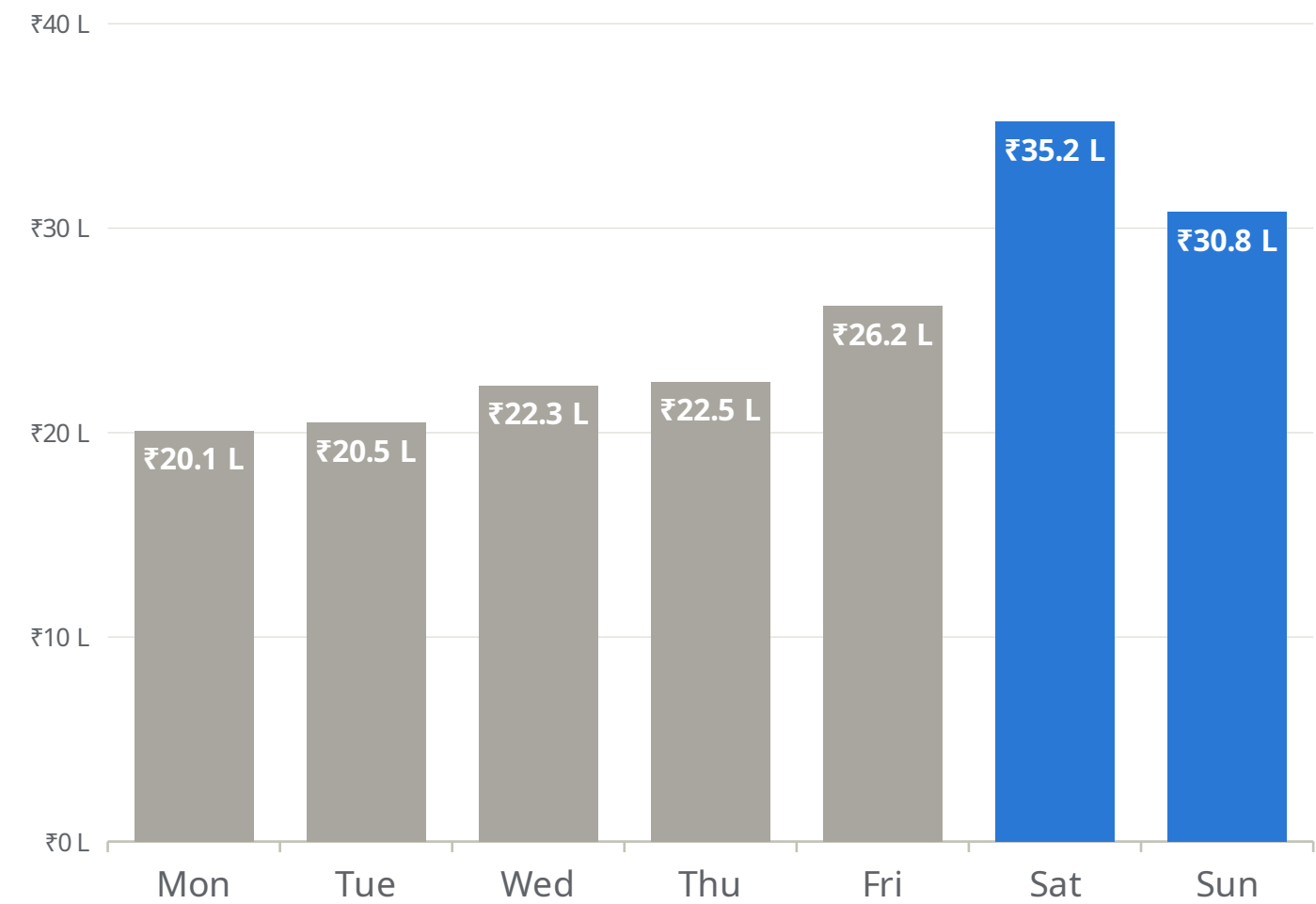
- Mobile Phone ranks 3rd of 5 by total revenue, so a top-2 / bottom-2 table misses it.
- ₹98.0 L in January → ₹42.1 L in June; transactions roughly halved (177 → 87).
- Electronics fell in all 6 cities — not confined to one store.

Top & bottom by total

Inside each category the five products earn almost the same (Grocery best vs worst: 2% apart). Direction matters more than the total.

Saturday brings in 75% more revenue than Monday

Average chain revenue per day (₹ Lakh) — weekends carry the week



Chain vs Apparel

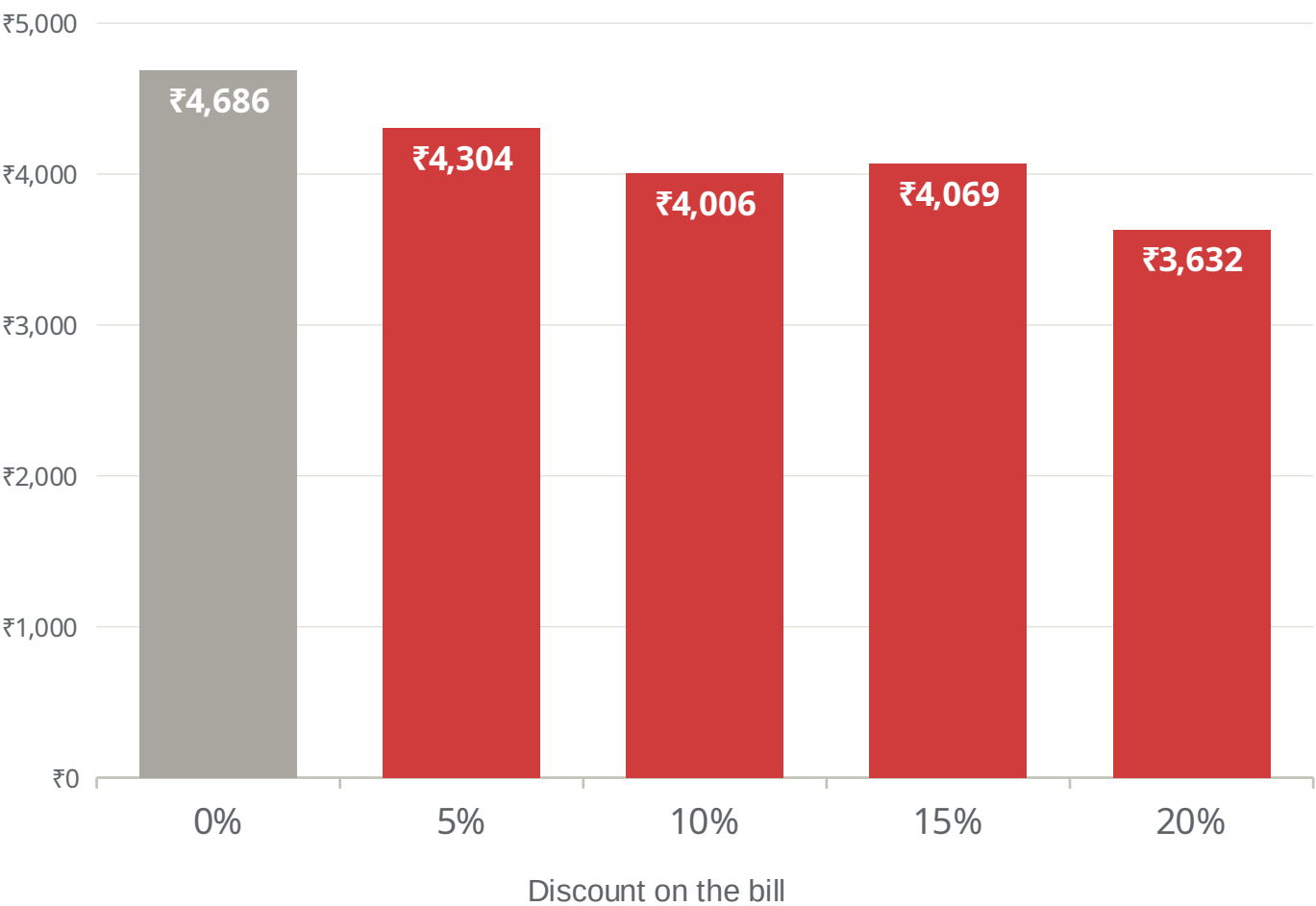
	Chain	Apparel
Best day	Saturday	Saturday
Worst day	Monday	Monday
Best month	January	June
Worst month	June	February

What it means for promotions

- Strong weekend sales suggest testing Monday–Thursday promotions; their added impact is unmeasured.
- Apparel grows overall despite a February dip (+34% vs chain –18%): prioritise stock over deeper discounts.
- Electronics: investigate range, availability and pricing — timing alone is unlikely to fix it.

Higher discounts: lower revenue, similar units

Revenue per transaction is generally lower at higher discounts (no transactions above 20%)



≈ 2.6

units per transaction at every discount level — 2.59 at full price, 2.58 at 20% off

–22%

revenue per transaction at 20% off (₹3,632 vs ₹4,686); similar pattern within every category

₹3.56 Cr

of discounts recorded in six months — ₹1.76 Cr of it on Electronics

Supports challenging deeper discounts; effect on total demand and profit is not established.

Our recommendation, category by category

Based on revenue and units — profit to be confirmed with margin data; the rules on the right are proposals to test



INVEST Apparel

₹6.95 Cr · +34%

More shelf space and stock depth — Jeans (+32% Q2 vs Q1) and T-Shirt (+30%) first.



CHALLENGE Electronics

₹22.65 Cr · -35%

Ask the supplier to explain range, availability and pricing. Reduce Mobile Phone and Tablet orders; keep Headphones.



WATCH Home & Kitchen

₹8.18 Cr · -18%

Reduce Dinner Set, Mixer Grinder, Water Bottle orders; keep Pressure Cooker and Storage Box.



HOLD Grocery

₹4.23 Cr · -4%

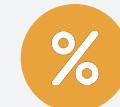
Half of all transactions; lowest revenue variability. No cuts; maintain availability.



HOLD Personal Care

₹3.96 Cr · +2%

Steady. Small increases; plan Sunscreen as seasonal.



Trial a 10% cap on store-funded discounts

A policy to test. Deeper discounts supplier-funded. Illustration: ≈ ₹83.7 L of discount not given in six months.



Test weekday promotions

₹1.32 Cr of discounts were recorded at weekends. Measure impact against control stores.



Apparel stock to the fastest-growing cities

Guntur (+30% units), Kakinada (+27%), Tirupati (+24%).

Category Manager Briefing

What we would say walking into the supplier review



Back Apparel

Revenue ₹1.01 Cr → ₹1.35 Cr (+34%), with all five products growing. More shelf space and stock.



Challenge Electronics

Start with Mobile Phones: ₹98.0 L → ₹42.1 L a month. Ask for an explanation before renewing shelf space and funding.



Challenge deeper discounts

20% off = ₹3,632 vs ₹4,686 per transaction, units ≈ unchanged. Suppliers fund deeper discounts; trial a 10% cap.



Data we need next

Product cost and margin — to confirm that moving shelf space to Apparel raises profit, not just revenue.

Reduce what is falling. Stock what is growing. Hold what is stable. Test — don't assume — the value of deeper discounts.